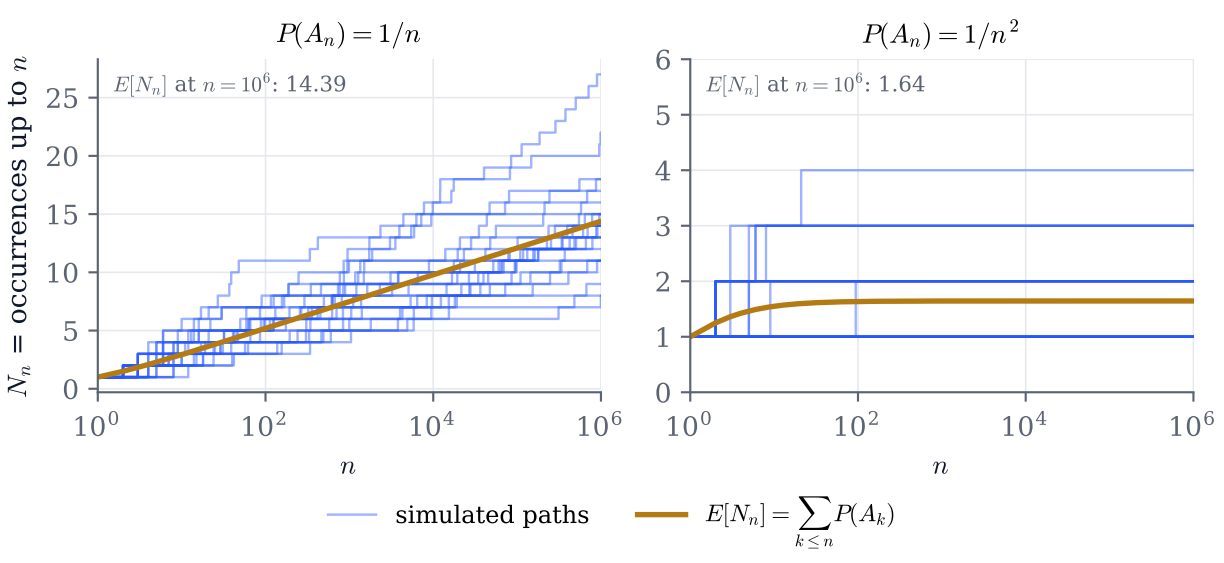
$P(A_n) = 1/n$
$E[N_n]$ at $n = 10^6$: 14.39
N_n = occurrences up to n
0
5
10
15
20
25
10^0
10^2
10^4
10^6
n
$P(A_n) = 1/n^2$
$E[N_n]$ at $n = 10^6$: 1.64
0
1
2
3
4
5
6
10^0
10^2
10^4
10^6
n
simulated paths
$E[N_n] = \sum_{k \le n} P(A_k)$